# Module-2

## Assignment

**Question:** What is Exploratory Testing?

**Answer:** Exploratory testing is an approach to software testing that is often described as simultaneous learning, test design, and execution. It focuses on discovery and relies on the guidance of the individual tester to uncover defects that are not easily covered in the scope of other tests.

**Question:** What is traceability matrix?

**Answer:** To protect against changes you should be able to trace back from every System component to the original requirement that caused its presence.

**Question:** What is Boundary value testing?

**Answer:** the goal of boundary value analysis is to find any issues which may arise due to incorrect assumptions about the system behavior. Testing these boundary values ensures that the software functions correctly. Every partition for boundary analysis has its minimum and maximum values**.**

**Question:** What is Equivalence partitioning testing?

**Answer:** Equivalence partitioning is a technique of software testing in which input data is divided into partitions of valid and invalid values, and it is mandatory that all partitions must exhibit the same behavior. If a condition of one partition is true, then the condition of another equal partition must also be true, and if a condition of one partition is false, then the condition of another equal partition must also be false. The principle of equivalence partitioning is, test cases should be designed to cover each partition at least once. Each value of every equal partition must exhibit the same behavior as other.

**Question:** What is Integration testing?

**Answer:** Testing performed to expose defects in the Interfaces and in the interactions between integrated components or systems.

lntegration Testing is a level of the software testing process where individual units are combined and tested as a group.

**Question:** What determines the level of risk?

**Answer:** Risks should be prioritized according to their level, which is obtained by assessing the likelihood of the event occurring and the impact of that event.

**Question:** What is Alpha testing?

**Answer:** It is always performed by the developers at the software development site.

- Sometimes it is also performed by Independent Testing Team.
- Alpha Testing is not open to the market and public
- It is conducted for the software application and project.
- It is always performed in **Virtual Environment**.
- It is always performed within the organization.
- It is the form of Acceptance Testing.
- Alpha Testing is definitely performed and carried out at the developing organizations location with the involvement of developers.
- It comes under the category of both White Box Testing and Black Box Testing.

**Question:** What is beta testing?

**Answer:** It is always performed by the customers at their own site.

- It is not performed by Independent Testing Team.
- Beta Testing is always open to the market and public.
- It is usually conducted for software product.
- It is performed in Real Time Environment.
- It is always performed outside the organization.
- It is also the form of Acceptance Testing.
- Beta Testing (field testing) is performed and carried out by users or you can say people at their own locations and site using customer data.
- It is only a kind of Black Box Testing.

**Question:** What is component testing?

**Answer**: Component (Unit) – A minimal software item that can be tested in isolation. It means "A unit is the smallest testable part of software."

- Component Testing – The testing of individual software components.

- Unit Testing is a level of the software testing process where individual units/components of a software/system are tested. The purpose is to validate that each unit of the software performs as designed.

**Question:** What is functional system testing?

**Answer:** A requirement that specifies a function that a system or system component must perform.

**Question:** What is Non-Functional Testing?

**Answer:** Testing the attributes of a component or system that do not relate to functionality, e.g.

reliability, efficiency, usability, interoperability, maintainability and portability.

**Question:** What is GUI Testing?

**Answer:** Graphical User Interface (GUI) testing is the process of testing the system's GUI of the System under Test. GUI testing involves checking the screens with the controls like menus, buttons, icons, and all types of bars – tool bar, Menu bar, dialog boxes and windows etc.

**Question:** What is Adhoc testing?

**Answer:** Adhoc testing is a software testing technique performed without any specific test plan or predefined set of steps.

**Question:** What is load testing?

**Answer:** Load Testing is done in order to check when the application fails by increasing the number of users and keeping the system resources as constant.

**Question**: What is stress testing?

**Answer:** Stress Testing is done in order to check when the application fails by reducing the system resources such as RAM, HDD etc. and keeping the number of users as constant.

**Question:** What is white box testing and list the types of white box testing?

**Answer:** Testing based on an analysis of the internal structure of the component or system. Structure-based testing technique is also known as 'white-box 'or 'glass-box' testing technique because here the testers require knowledge of how the software is implemented, how it works.

**Types of white box testing:**
Web Based Testing
Desktop Based Testing
Mobile Based Testing
Game Based Testing

**Question:** What is black box testing? What are the different black box testing techniques?

**Answer:** Testing, either functional or non-functional, without reference to the internal structure of the component or system.

Specification-based testing technique is also known as 'black-box 'or input/output driven testing techniques because they view the software as a black-box with inputs and outputs.

Black box testing techniques: Equivalence partitioning, Boundary value analysis, Decision tables, State transition testing, Use-case Testing.

**Question:** Mention what big bang testing is?

**Answer:** In Big Bang integration testing all components or modules is integrated simultaneously, after which everything is tested as a whole.

Big Bang testing has the advantage that everything is finished before integration testing starts.

The major disadvantage is that in general it is time consuming and difficult to trace the cause of failures because of this late integration.

Here all component are integrated together at once, and then tested.

**Question:** What is the purpose of exit criteria?

**Answer:**

- Successful Testing of Integrated Application.
- Executed Test Cases are documented.
- All High prioritized bugs fixed and closed.
- Technical documents to be submitted followed by release Notes.

**Question:** When should "Regression Testing" be performed?

**Answer:** The purpose of regression testing is to confirm that a recent program or code change has not adversely affected existing features.
Regression testing is nothing but full or partial selection of already executed test cases which are re-executed to ensure existing functionalities work fine.
This testing is done to make sure that new code changes should not have side effects on the existing functionalities. It ensures that old code still works once the new code changes are done.

**Question:** What is 7 key principles? Explain in detail?

**Answer:**

1. Testing shows presence of Defects

2. Exhaustive Testing is Impossible!

3. Early Testing

4. Defect Clustering

5. The Pesticide Paradox

6. Testing is Context Dependent

7. Absence of Errors Fallacy

**Question:** Difference between QA v/s QC v/s Tester

**Answer: Quality Assurance**

- Activities which ensure the implementation of processes,
- Procedures and standards in context to verification of developed software and intended requirements.
- Focuses on processes and procedures rather than conducting actual testing on the system.
- Process oriented activities.
- Preventive activities.
- It is a subset of Software Test Life Cycle (STLC).

**Quality Control**

- Activities which ensure the verification of developed software with respect to documented (or not in some cases) requirements.
- Focuses on actual testing by executing Software with intend to identify bug/defect through implementation of procedures and process.
- Product oriented activities.
- It is a corrective process.
- QC can be considered as the subset of Quality Assurance.

**Tester or Testing**

- Activities which ensure the identification of bugs/error/defects in the Software.
- Focuses on actual testing.
- Product oriented activities.
- It is a preventive process.
- Testing is the subset of Quality Control.

**Question:** Difference between Smoke and Sanity?

**Answer: Smoke Testing**

- Smoke Testing is performed after software build to ascertain that the critical functionalities of the program is working fine.
- It is executed "before" any detailed functional or regression tests are executed on the software build.
- The purpose is to reject a badly broken application, so that the QA team does not waste time installing and testing the software application.
- In Smoke Testing, the test cases chosen cover the most important functionality or component of the system.
- The objective is not to perform exhaustive testing, but to verify that the critical functionalities of the system are working fine.

**Sanity Testing**

- After receiving a software build, with minor changes in code, or functionality, Sanity testing is performed to ascertain that the bugs have been fixed and no further issues are introduced due to these changes.
- The goal is to determine that the proposed functionality works roughly as expected.
- If sanity test fails, the build is rejected to save the time and costs involved in a more rigorous testing.
- The objective is "not" to verify thoroughly the new functionality, but to determine that the developer has applied some rationality (sanity) while producing the software.

**Question:** Difference between verification and Validation

**Answer:** Validation is the process of checking whether the specification captures the customer's requirements, while verification is the process of checking that the software meets specification.

**Question:** Explain types of Performance testing.

**Answer:** Load testing, Stress testing, Endurance testing, Spike testing, Volume testing

Scalability testing.

**Question:** What is Error, Defect, Bug and failure?

**Answer:**

**Failure:** The inability of a system or component to perform its required functions within specified performance requirements. See: bug, crash, exception, and fault.

**Bug:** A fault in a program which causes the program to perform in an unintended or unanticipated manner. See: anomaly, defect, error, exception, and fault. Bug is terminology of Tester.

**Fault:** An incorrect step, process, or data definition in a computer program which causes the program to perform in an unintended or unanticipated manner. See: bug, defect, error, exception.

A mistake made by a programmer during coding is called an error, an error found during the unit testing in the development phase is called defect, an error found during the testing phase is called bug and when an error is found at an end user's end is called as the failure.

**Question:** Explain the difference between Functional testing and Nonfunctional testing?

**Answer:**

Functional requirements define how the system must work and non-functional requirements detail how it should perform. Without functional requirements the system will not work. Without at least some non-functional requirements being met to a certain level, users will become frustrated.

**Question:** What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)?
**Answer:** SDLC covers the entire software development process, including planning, analysis, design, coding, testing, deployment, and maintenance. STLC is a part of SDLC and focuses specifically on testing. STLC includes activities like test planning, test case development, test execution, defect tracking, and closure.

**Question:** What is the difference between test scenarios, test cases, and test script?

**Answer:** A test case is a document with instructions on testing the specific functionality of an application. Test Script is a program that runs various test data on the functionality of an application. Test scenarios serve as an outline for writing test cases.

**Question:** Explain what Test Plan is? What is the information that should be covered?

**Answer:** A test plan is a detailed and comprehensive software testing document that outlines the strategy, objectives, schedule, resources, and overall approach that are needed to complete project.

**Question**: What are the different Methodologies in Agile Development Model?

**Answer:** The Agile methodology is a way to manage a project by breaking it up into several phases. It involves constant collaboration with stakeholders and continuous improvement at every stage. Once the work begins, teams cycle through a process of planning, executing, and evaluating.

Agile methodology is the type of methodology in which a user can see something quickly. It is the iterative as well as its incremental process. It prioritizes user collaboration, responding to change and delivering the software product in short time.

**Question**: Explain the difference between Authorization and Authentication in Web testing. What are the common problems faced in Web testing?

**Answer:** Authentication and Authorization are both utilized in data security, allowing the safeguarding of an automated data system. Both are very crucial topics often associated with the internet as key components of its service infrastructure. However, each term is distinct, representing different concepts. While they are frequently used in the same context with the same tools, they are entirely distinct from one another. In the authentication process, users' identities are verified to grant access to the system, whereas in the authorization process, a person's or user's permissions are verified to access resources. Authentication precedes the authorization process, while the authorization process occurs after authentication.

**Question:** When to used usability testing?

**Answer:** The goal of usability testing is to understand how real users interact with your website and make changes based on the results.

Usability testing is typically conducted at various stages of the design process, from initial concept to post-launch evaluation. It involves asking participants to perform specific tasks related to the product or service while their interactions and feedback are observed and recorded.

**Question:** What is the procedure for GUI Testing?

**Answer:** A GUI uses windows, icons, and menus to carry out commands, such as opening files, deleting files and moving files. GUI software is much easier to use than command line software since there is no need to type in and memorize individual commands.